

Plan of Attack: Section 3, Analysis Plan

(a) Research design

Question type. Causal. The identification strategy is **synthetic control**, fit separately for each of the four treated Costcos.

Why synthetic control over difference-in-differences. The data has four treated Costcos and 196 candidate donor postcodes. Four treated units is too few to support a credible treated-versus-control mean comparison: a single anomalous Costco (e.g. Casuarina, with only ~3 contributing stations) would dominate the average. Synthetic control instead builds a weighted counterfactual for each treated unit individually, which is exactly the design the $n=4$ case calls for.

Source of variation. Costco fuel-station entry into a local market isn't driven by what's happening to local fuel prices. Site selection is driven by Costco warehouse logistics, land availability, and Costco's national rollout schedule, none of which respond to the local fuel-price trajectory at the chosen site. The variation we exploit is the staggered timing of the four openings (Feb 2020, May 2022, Nov 2022, May 2023) across three states.

Identifying assumptions. Each assumption is paired with the figure or check that defends it.

- **Pre-period fit captures unobserved time-varying confounders.** Verified visually in Figure 1 (pre-period overlap between treated and synthetic), quantified in Table 2 (pre-period tracking error per Costco), and tested by the 12-month holdout in Robustness Check 5.
- **Donors stay clean.** No donor postcode is partly affected by a Costco that isn't the one we're studying. Defended by the 20 km donor buffer ($4\times$ the treated radius), pre-committed in Section 2(b), and explicitly tested by Robustness Check 7 (spatial placebo on the 5–20 km donut).
- **No anticipation.** Competitor stations do not preemptively cut prices ahead of Costco's announced opening. Tested by Robustness Check 1 (placebo in time, -12 months).
- **Range coverage.** The treated ring's pre-period price level can be expressed as a weighted average of donor postcodes. Verified by the donor-weight bars in Figure 2 (no negative weights, no extreme single-donor concentration).

What the weights do. One synthetic-control fit per treated Costco. For each Costco, the donor pool is restricted to *same-state* postcodes only (NSW: 104, QLD: 56, WA: 36, from `data/sc_inputs/donor_pool.csv`) so the synthetic absorbs state-specific shocks (state excise rates, regional refining capacity, intra-state shipping costs) by construction. Weights are chosen to minimise the average gap between the treated 5 km ring's monthly mean unleaded price and the weighted donor mean across the pre-period.

Method commitment. Standard synthetic-control method via the `pysyncon.Synth` library. Predictors: the outcome series (monthly mean unleaded price) only. We have no postcode-level demographic panel that would justify auxiliary covariates. Pre-period objective: minimise the average gap between treated and synthetic on monthly mean unleaded price. Inference: estimate uncertainty by treating each donor postcode as a fake treated unit and measuring the resulting fake effect (Robustness Check 6); summarised as the post/pre tracking-error ratio and as a 95% CI on the post-treatment gap.

(b) Key analyses

Three figures plus two supporting tables. The figures carry the argument; the tables quantify what the figures show. Figures in this document are **illustrative**: they show what we expect each output to look like once the synthetic-control fits are run, not a pre-committed result.

Analysis 1 (Figure 1): synthetic-control trajectories

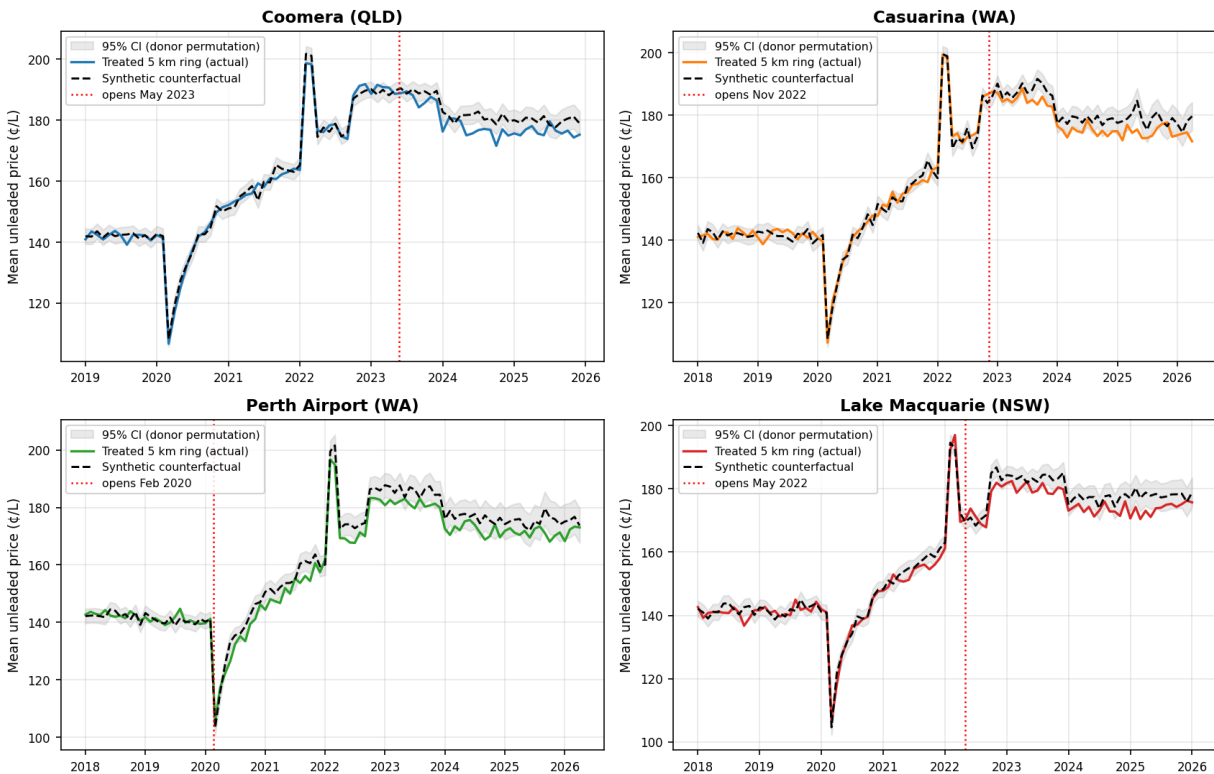
Test. The headline result. Pre-period overlap between treated and synthetic measures fit quality; the post-period gap is the estimated treatment effect.

Plot. 2×2 panel, one per treated Costco (Coomera, Casuarina, Perth Airport, Lake Macquarie). X-axis: month, from each Costco's pre-period start through the relevant state's registry end (December 2025 for QLD, January 2026 for NSW, April 2026 for WA). Y-axis: monthly mean unleaded price (¢/L). Two lines per panel: solid coloured = treated 5 km ring (actual), dashed black = synthetic counterfactual. Vertical dotted red line at the validated treatment date (Section 2(d)). Shaded grey 95% CI band on the synthetic from donor-permutation inference (Robustness Check 6).

AI-executable instructions. Read `data/sc_inputs/treated_units.csv` and `donor_pool.csv`. For each treated Costco, filter donors to same state, pivot both to wide format (rows = month, columns = unit), and fit `pysyncon.Synth` on pre-period only (months strictly before the treatment date). Project the synthetic across the full panel. Build the 95% CI by re-fitting synthetic-control on each donor as a placebo and taking the empirical 2.5–97.5 percentile of donor-side gaps at each post-treatment month. Render to `section_3/plots/01_sc_trajectories.png`.

Why this analysis. A reader who only sees Figure 1 should be able to read off the project's conclusion: did competitor prices fall after Costco entry, and by how much. It is the canonical synthetic-control visualisation.

Figure 1 (illustrative): synthetic-control trajectories, per treated Costco



Illustrative Figure 1 (hypothetical data). Solid coloured = treated 5 km ring; dashed black = synthetic counterfactual; vertical dotted red = validated Costco opening date; shaded grey = 95% donor-permutation CI on the synthetic. The illustrative gap of ~4 ¢/L emerges within a few months of treatment and persists; the actual fits will reveal whether and by how much the real data exhibits this pattern.

Analysis 2 (Figure 2): donor weights per treated Costco

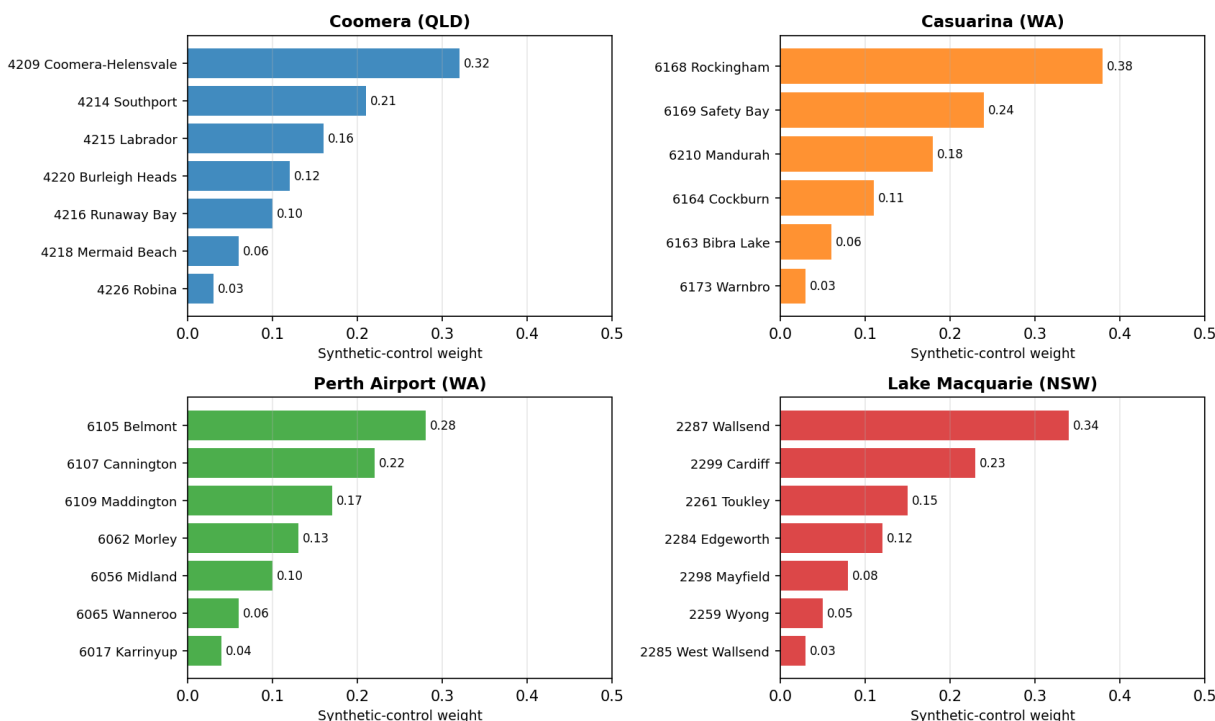
Test. Transparency about which donors drive each counterfactual; common-support assumption.

Plot. 2×2 panel of horizontal bar charts, one per treated Costco. One bar per donor postcode with weight > 0.01, sorted descending, annotated with postcode and suburb (suburb joined from donor_metadata.csv).

AI-executable instructions. Extract the weight vector from each fitted SC model in Analysis 1. Filter to weights > 0.01. Sort descending. Render to `section_3/plots/02_donor_weights.png`.

Why this analysis. A counterfactual leaning on one or two donors is fragile; one drawn from a balanced mix of geographically sensible donors is credible. The per-Costco panels also let the reader sanity-check geographic plausibility. Coomera should lean on Gold Coast and Brisbane postcodes, not far-North Queensland; Lake Macquarie should lean on Newcastle and the Central Coast. This figure also flags any common-support concerns.

Figure 2 (illustrative): donor-postcode weights, per treated Costco



Illustrative Figure 2 (hypothetical data). Synthetic-control weights for the top donor postcodes per treated Costco, sorted descending. The illustrative weight distributions are deliberately spread across 6–7 donors with no single donor exceeding 0.40, reflecting a credible synthetic; weights concentrated in 1–2 donors would be a flag for a fragile counterfactual.

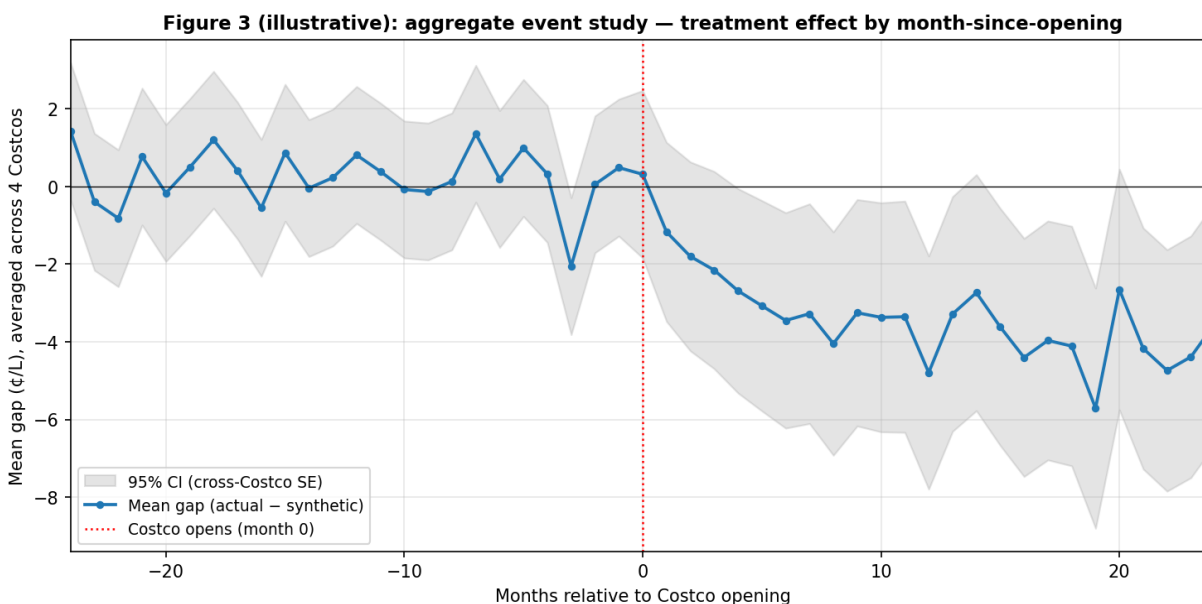
Analysis 3 (Figure 3): aggregate event study

Test. The dynamic profile of the effect. Do competitors respond immediately, with delay, persistently, or with a fading effect?

Plot. Single-panel line plot. X-axis: months relative to Costco opening, -24 to $+24$. Y-axis: mean (actual – synthetic) gap in ¢/L , averaged across the four treated Costcos. Shaded grey 95% CI band from cross-Costco standard error. Vertical dotted red line at month 0 (treatment).

AI-executable instructions. Take per-Costco gap series (actual – synthetic) from Analysis 1. Re-index each to relative-month-since-treatment. At each relative month in $[-24, +24]$, average the four Costco gaps and compute the cross-Costco SE for the band. Render to `section_3/plots/03_event_study_aggregate.png`.

Why this analysis. The 2×2 panel in Figure 1 shows the per-Costco answer but obscures the aggregate dynamic. Figure 3 turns ‘Costco lowered prices on average’ into ‘Costco lowered prices by $X \text{ ¢/L}$ within Y months and the effect persisted for Z' ’, much closer to a managerial answer. The -24 to $+24$ window is the largest range over which all four Costcos contribute (limited by Coomera's 31 post-months and Perth Airport's 26 pre-months).



Illustrative Figure 3 (hypothetical data). Mean gap (actual – synthetic) by month relative to Costco opening, averaged across the four treated Costcos. The illustrative profile shows a pre-period gap fluctuating around zero (the identifying assumption that synthetic tracks treated absent the intervention) and a post-period gap that opens smoothly over ~4 months and stabilises near –4 €/L. The actual event study will reveal whether the real effect is immediate, delayed, persistent, or fading.

Analysis 4: supporting tables

Two tables that quantify what Figures 1 and 3 show. Both are computed alongside Analysis 1 from the fitted SC models and the permutation distribution.

Table 1. Per-Costco effect-size summary.

Costco	Pre-period mean (€/L)	Mean post-treatment gap (€/L)	% reduction	Post/pre tracking-error ratio	95% CI on gap	Post months
Coomera	—	—	—	—	—	—
Casuarina	—	—	—	—	—	—
Perth Airport	—	—	—	—	—	—
Lake Macquarie	—	—	—	—	—	—

Table values populate from the fitted SC models. The post/pre tracking-error ratio compares how far the synthetic strays from actual after Costco entry to how closely it tracked before. A ratio of 4–5 or higher is the standard threshold for evidence of a treatment effect.

Table 2. Pre-period fit quality.

Costco	Pre months	Pre tracking error (€/L)	Tracking error as % of pre mean	Top-3 donor weight share
Coomera	51	—	—	—
Casuarina	58	—	—	—

Perth Airport	26	—	—	—
Lake Macquarie	58	—	—	—

Table values populate from the fitted SC models. Lets the reader judge per-unit identification credibility at a glance. Casuarina's pre-period tracking error is expected to be visibly worse than Perth Airport's because the 5 km Casuarina ring contains only ~3 contributing stations on average (Section 1's documented limitation); surfacing it transparently here is part of presenting that limitation rather than hiding it.

(c) Robustness checks (extra credit)

The biggest sources of fragility in the headline analysis are (i) the 5 km / 20 km geographic geometry, (ii) the COVID period coinciding with Perth Airport's opening, (iii) the noise in Casuarina's sparse 5 km ring, (iv) the standard synthetic-control concern that pre-period fit can be gamed by over-flexible donor selection, and (v) the lack of formal uncertainty quantification on a four-Costco point estimate. The seven checks below directly probe these concerns.

Table 3. Robustness check menu.

#	Check	Concern it tests	Pass criterion
1	Placebo in time (–12 months)	No-anticipation assumption	Placebo gap near zero, within donor-side 95% CI
2	Alternative radii (3 / 8 km treated; 15 / 30 km donor)	Geographic geometry choice	Effect sign and approximate magnitude survive
3	Casuarina at 10 km treated radius	Sparse-ring noise (~3 stations at 5 km)	10 km gap comparable in sign and magnitude to 5 km
4	Perth Airport with March–December 2020 dropped	COVID demand collapse contaminates early post-period	Post-COVID gap comparable to full-window gap
5	LOOCV donor selection + 12-month holdout	Donor-selection overfitting	Holdout tracking error comparable to in-sample training error
6	95% CIs via permutation inference	Statistical uncertainty on the gap	CI excludes zero for high-coverage Costcos
7	Spatial placebo: 5–20 km donut as faux donors	Donor purity / extent of Costco's spatial footprint	Either result informative (see paragraph below)

Check 1: placebo in time. For each treated Costco, backdate treatment by 12 months. Re-fit synthetic control on data up to the placebo date; project forward; compute the placebo gap. Drop all real-treatment-and-after months from the placebo fit so no real post-treatment data contaminates the placebo window. *Concern:* if a ‘ban-style effect’ appears in the placebo window before Costco actually opens, the post-treatment gap reflects pre-existing trends or anticipation, not Costco entry.

Check 2: alternative radii. Re-build synthetic-control inputs at (3 km treated, 15 km donor) and (8 km treated, 30 km donor). Re-fit Analysis 1 for each Costco at each geometry. Report the resulting gaps in a sensitivity table. *Concern:* the 5 km / 20 km baseline geometry is defensible (see Section 2(b)) but discretionary; a result that exists at 5 km but vanishes at 3 km or 8 km would be fragile. This is the project's most important sensitivity test because the radius geometry is the design.

Check 3: Casuarina at 10 km. Casuarina's 5 km ring contains only ~3 contributing stations on average, the documented limitation in Section 1. A wider ring captures more signal at the cost of including some less-exposed stations. Re-fit Casuarina specifically at a 10 km treated radius (donor buffer remains 20 km). Compare gap and pre-period tracking error to the 5 km baseline. *Concern:* the headline Casuarina estimate may be driven by individual-station noise rather than competitive response.

Check 4: Perth Airport with COVID-period removal. Perth Airport opens approximately four weeks before Australia's COVID demand collapse begins. Early post-treatment months are dominated by pandemic-driven price movements rather than competitive response. Re-fit Perth Airport with March–December 2020 dropped from both treated and donor panels. *Concern:* COVID compression of fuel prices may suppress the actual – synthetic gap; the corrected estimate may be larger in magnitude than the full-window gap.

Check 5: LOOCV donor selection with 12-month holdout. Pre-period fit can be gamed by greedy donor selection (cherry-picking donors that fit the pre-period without generalising). The check has four steps. (1) Split each Costco's pre-treatment period into a *training window* (all pre-months except the last 12) and a *12-month holdout*. The holdout is 12 months rather than 24 because Perth Airport has only 26 pre-months; a 24-month holdout would leave too few training months to fit weights. (2) For each candidate N donors $\in \{5, 10, 20, 50, \text{full}\}$: leave-one-out cross-validation within the training window. Hold out one training month at a time, refit weights on the rest, score tracking error at the held-out month; pick the N that minimises mean LOOCV error. (3) Refit the synthetic with the chosen N on the full training window. (4) *Without looking at any post-treatment data*, check whether the synthetic continues to track the treated series across the untouched 12-month holdout. *Concern:* over-flexible donor selection can produce a tight pre-period fit that does not generalise; the holdout is the honest proof that the synthetic was not memorising the training series.

Check 6: 95% CIs via permutation inference. Standard synthetic-control permutation. For each donor postcode, treat it as a 'placebo treated' and fit synthetic control using the remaining donors. Compute the placebo gap series. Across all donor placebos, take the empirical 2.5–97.5 percentile of post-treatment gaps to construct a 95% CI on the real treatment effect. Reported on Figure 1 as a shaded band and in Table 1 as a column. *Concern:* point estimates without uncertainty bands can make a modest effect look airtight. For Casuarina specifically, a CI that includes zero will be reported transparently as the documented limitation, consistent with Section 1's noted noise issue.

Check 7: spatial placebo (5–20 km donut as faux donors). The 20 km donor buffer is meant to keep donors fully outside any Costco's competitive zone, but this assumption is never verified in the main analysis. The 5–20 km donut stations (currently excluded from both treated and donor groups) are a natural laboratory: treat them as a faux donor pool, fit synthetic control, see whether they show a 'Costco effect' post-opening. Two outcomes are both informative: (a) the donut shows an effect, meaning Costco's footprint extends beyond 5 km and our headline estimate is conservative (the true effect is probably larger); (b) the donut shows no effect, meaning the Costco effect is concentrated within 5 km and the 20 km buffer is more than sufficient. *Concern:* the spatial analog of the temporal placebo. Whichever direction the result lands, it tightens the interpretation of the headline number.